

Lecture 7: Inverse problems and the story of DNA

Today's plan

An overview of inverse problem

The story of the discovery of DNA

What is an inverse problem?

Usually in mathematics classes we ask you to find a solution to an equation

$$\frac{d^2 x}{dt^2} + x = 0$$

$$ax^2 + bx + c = 0$$

$$a^n + b^n = c^n$$

$$\frac{\partial c}{\partial t} = \mathcal{D} \nabla^2 c$$

For an inverse problem, we give you the solution to an equation and ask you to find the equation

Not a well-defined question, but understanding limitations often leads to new mathematics!

Example: multiplication

- **Forward problem: multiply two integers a and b to get n**
- **Inverse problem: given an integer n , factorize it into a and b**
- **Inverse problem does not have a unique solution (*e.g.* $24 = 8*3 = 4*6 = 12*2 = 1*24$)**
- **However, this also leads to interesting mathematics by itself (*e.g.* prime numbers)**

Examples

- Find a polynomial of degree n which has a root at $x = 2$
- Find a differential equation whose solution is $x(t) = e^t$
- Given two points in a plane, find a circle that passes through the two points

Examples

- **Can you solve for the integrand?**

$$\int_0^1 f(x) dx = 1$$

- **What if the integrand attains its maximum at $x = 1/2$?**

Sequences: what rule generates the following sequences?

1, 2, 4, 8, 16, 32, 64, 128, 256, ...

Powers of 2: $a_n = 2^n$

1, 3, 6, 10, 15, 21, 28, 36, 45, 55, ...

Triangular numbers: $T_n = \frac{n(n+1)}{2}$

Practical example: GPS

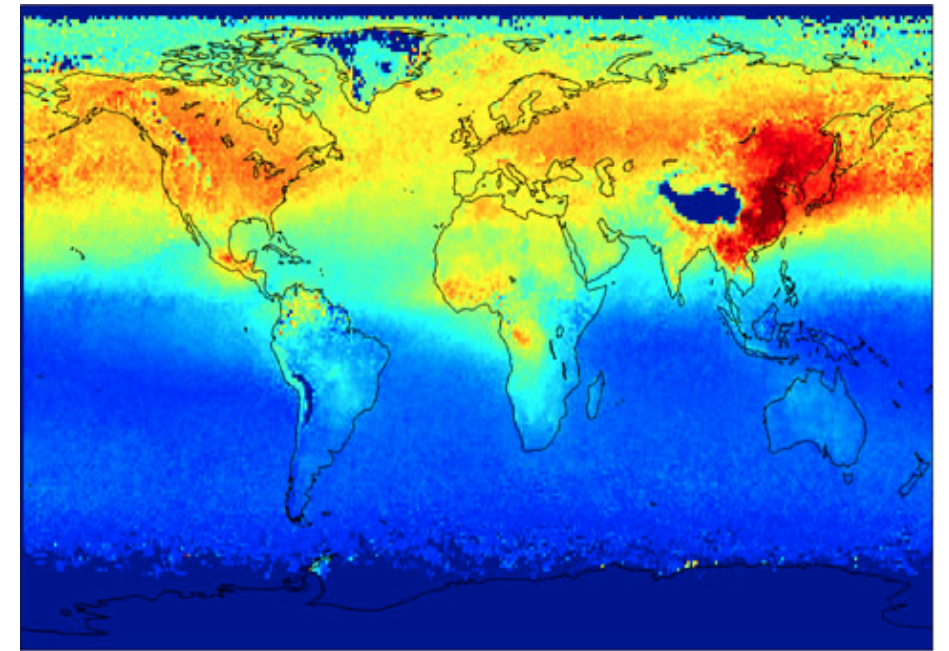
- **Measure the distance to each satellite**
- **Given the distance, know that the receiver lies on a sphere centered at the satellite**
- **Each satellite corresponds to a different sphere**
- **Receiver is located at the intersection of a point of the spheres**
- **How many are needed?**



More practical examples

Atmospheric pollution monitoring

Where is the pollution originating, given satellite measurements of chemical composition?



The flying insect

Can smell food but cannot see it.
How should it move to find the source of food?



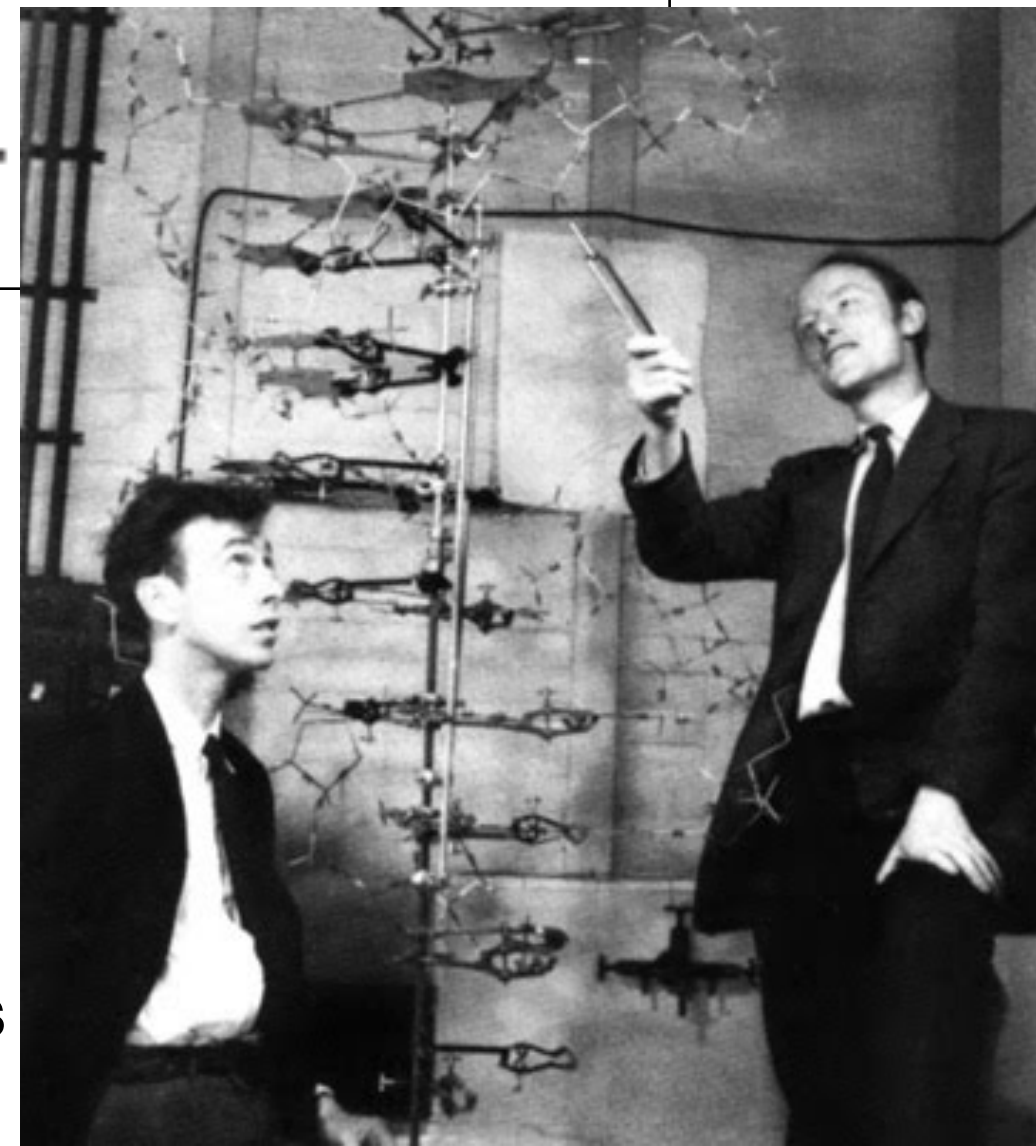
Break

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

J. D. WATSON
F. H. C. CRICK

Medical Research Council Unit for the
Study of the Molecular Structure of
Biological Systems,
Cavendish Laboratory, Cambridge.
April 2.



**1952 Nobel Prize for Physiology and Medicine
James Watson, Francis Crick, and Maurice Wilkins**

Motivation

- 1944 paper by Avery, MacLeod and McCarty shows that DNA is the material of inheritance
- Watson and Crick realized from model building that A–T and C–G have the same area and shape
- Early 1950's: what structure could support these units?

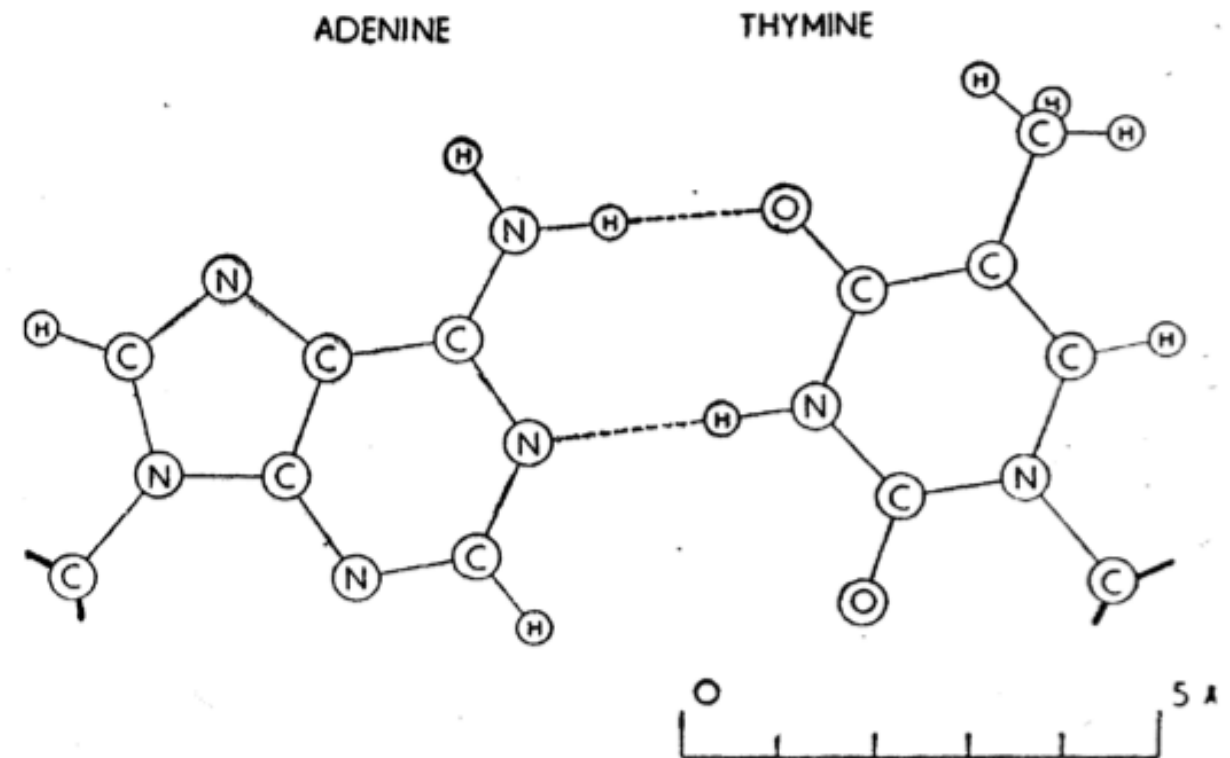


Fig. 4. Pairing of adenine and thymine. Hydrogen bonds are shown dotted. One carbon atom of each sugar is shown

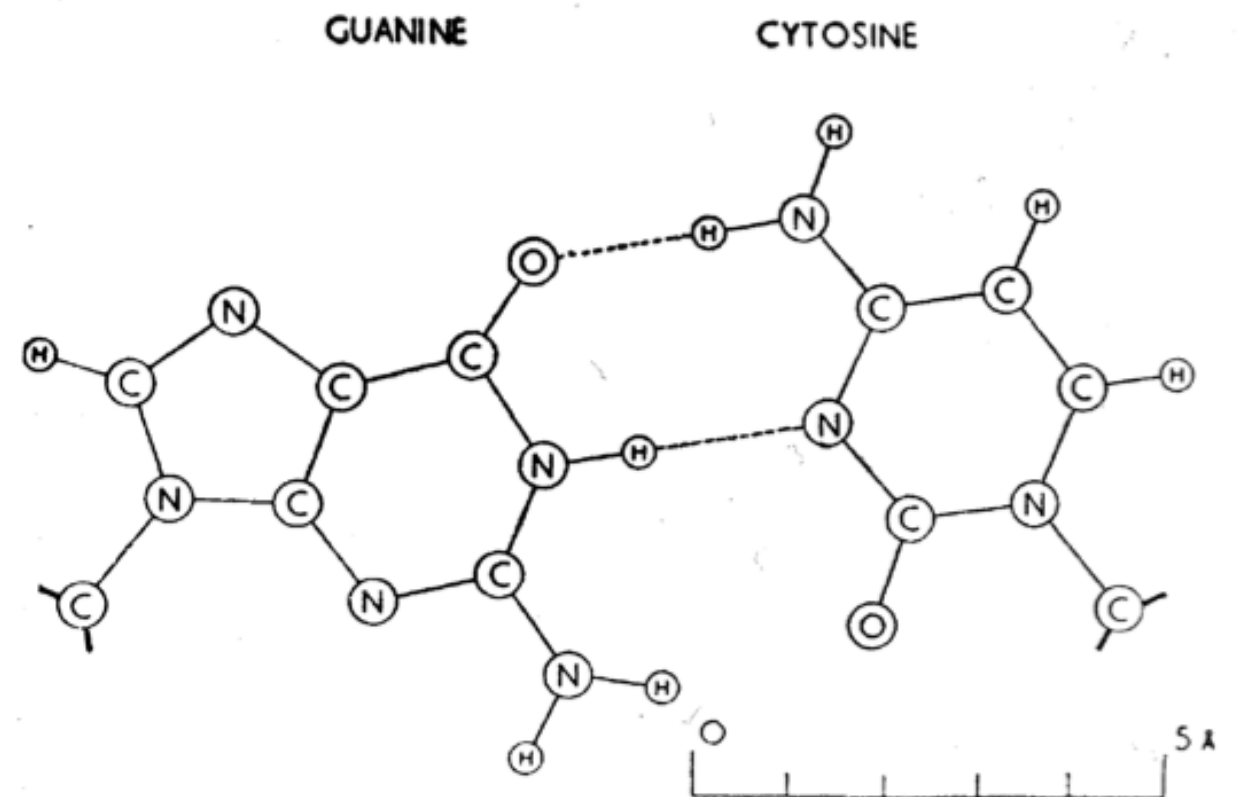
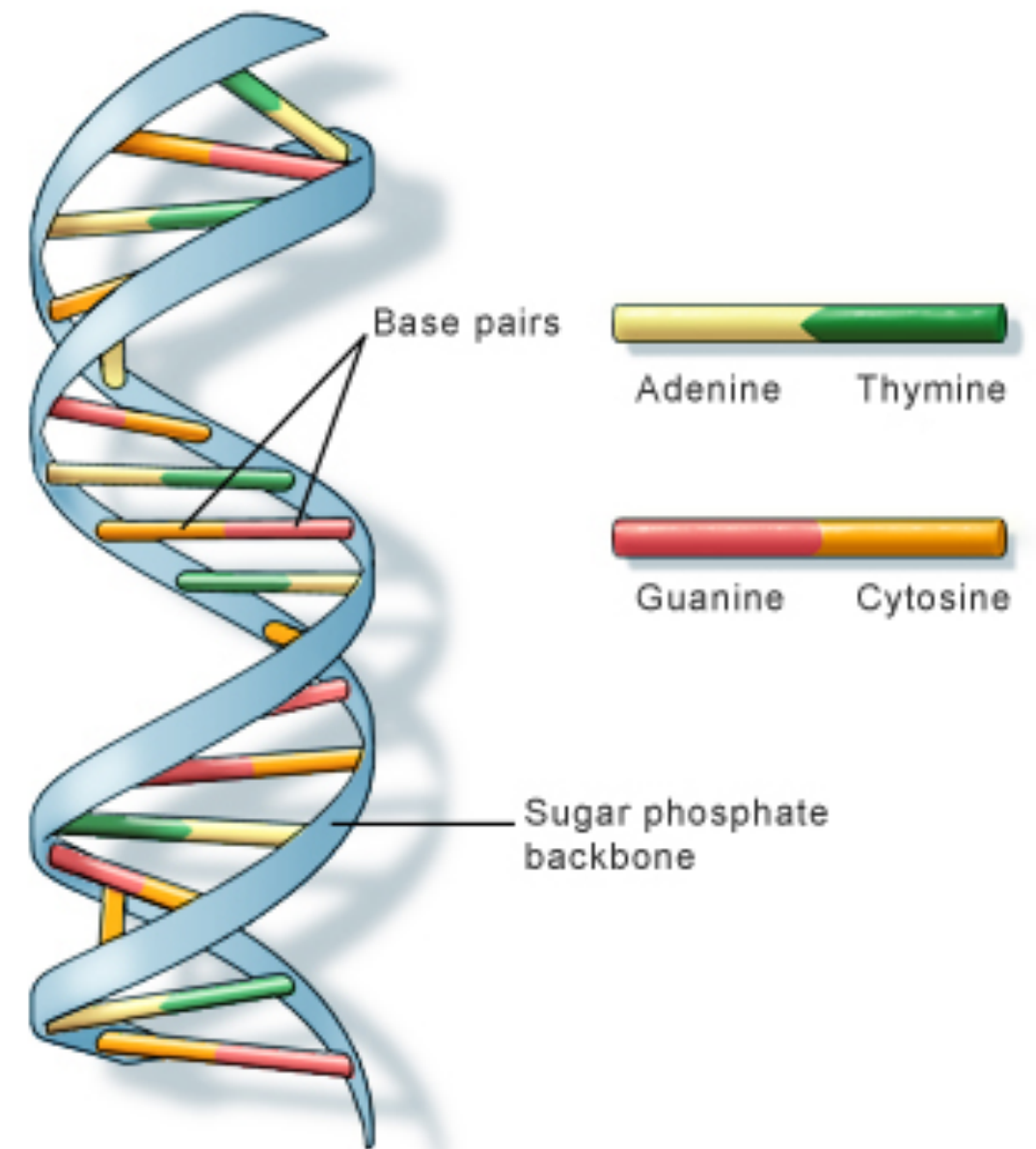


Fig. 5. Pairing of guanine and cytosine. Hydrogen bonds are shown dotted. One carbon atom of each sugar is shown

What structure?



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X-ray diffraction

- **It is not possible to see DNA directly; it is too small.**
- **To see something, the wavelength of light must be much smaller than the size of the object. This is not true with DNA.**
- **But it is possible to see the diffraction pattern made by DNA using light rays with very small wavelength—X-rays.**

Rosalind Franklin and the Discovery of the Structure of DNA

by

A. KLUG

MRC Laboratory of Molecular Biology,
Cambridge

In this article Dr Klug discusses Dr Franklin's contribution to the discovery of the structure of DNA in the light of accounts given by Professor Watson in his book *The Double Helix* and by Dr Hamilton in a recent article in *Nature*.

ROSALIND FRANKLIN made crucial contributions to the solution of the structure of DNA. She discovered the B form, recognized that two states of the DNA molecule existed and defined conditions for the transition. From early on, she realized that any correct model must have the phosphate groups on the outside of the molecule. She laid the basis for the quantitative study of the diffraction patterns, and after the formulation of the Watson-Crick model she demonstrated that a double helix was consistent with the X-ray patterns of both the A and B forms.

Watson's account in *The Double Helix* does not pretend to tell more than one side of the story. The article by Dr L. D. Hamilton ("DNA: Models and Reality", *Nature*, May 18, 1968) does not do justice to Franklin's work*.

The importance of Franklin's work has been lost sight of, partly because of her untimely death. Because, as her last and perhaps closest scientific colleague, I am in a position to fill in the record, I have endeavoured here to

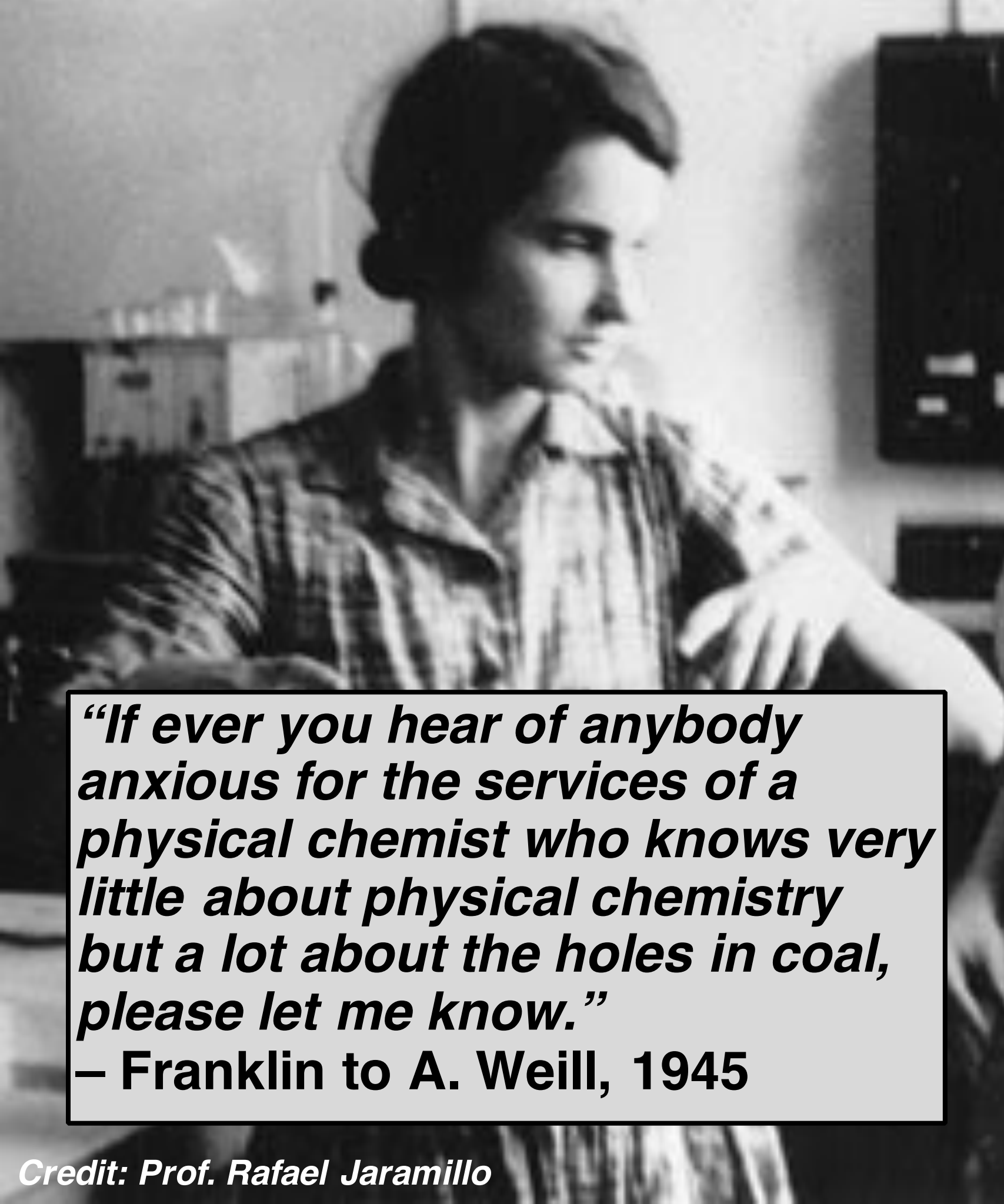
* For example, although both the A and B types of X-ray patterns given by DNA fibres are discussed, it is not stated that it was Franklin who discovered the B structure and also took the particular photograph referred to by Watson in the passages quoted by Hamilton. Likewise her role in demonstrating the validity of the Watson-Crick model for both the B and A forms—once it had been proposed—is not brought out: ref. 4 is omitted. Hamilton refers, like Watson, to her "anti-helical" view, a term which does not fairly reflect her attitude from about the end of 1952 onwards. This might be more accurately described as one of questioning; the question being whether the structure of B—undoubtedly helical in her view—also applied to the crystalline structure A.

give an account of what Franklin was doing in the period before the discovery of the Watson-Crick model, to place the helical question in context, and to summarize the contributions she made to the proof of the structure. I have not attempted to deal with the well recognized contributions made by the other protagonists in the story except in so far as they touch directly on her work.

The Published Record

Rosalind Franklin published, together with her student, R. G. Gosling, five papers on DNA¹⁻⁵. Much of the material in these papers is not readily intelligible to the non-crystallographer, which may account for the lack of attention they have received. I have used chiefly these published sources, supplemented by reference to Gosling's thesis⁶ and to Franklin's notebooks and reports which passed to me on her death in 1958.

The first two papers^{1,2} were sent to press in March 1953 before Franklin knew of the Watson-Crick model. The first¹ describes the observations on the types of X-ray diagram given by highly orientated specimens of sodium DNA at different humidities. Two forms of DNA fibres, named A and B, are described and the conditions are given for producing them. In this paper are reproduced the beautiful X-ray photographs which were used in the subsequent analysis of both forms (Fig. 1). The accompanying paper in *Acta Crystallographica* describes quanti-



***“If ever you hear of anybody
anxious for the services of a
physical chemist who knows very
little about physical chemistry
but a lot about the holes in coal,
please let me know.”***

– Franklin to A. Weill, 1945

Birkbeck
College

King's
College
Paris

University of
Cambridge

St. Paul's
School

1958

DNA papers in
Nature

Pauling's helical
protein

Schrodinger
What is Life?

Battle of Britain

1920

Bragg

Randall

***“This means that as far as the
experimental x-ray effort is
concerned there will be at the
moment only yourself and
me.”
Letter to Franklin, 1951***

STOP

Schrodinger

Peru

Gosling

Pauling

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

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April 2.

We have also been stimulated by
a knowledge of the general nature of the unpublished
experimental results and ideas of Dr. M. H. F.
Wilkins, Dr. R. E. Franklin and their co-workers at
King's College, London.

Molecular Structure of Deoxypentose Nucleic Acids

M. H. F. WILKINS

Medical Research Council Biophysics
Research Unit,

A. R. STOKES
H. R. WILSON

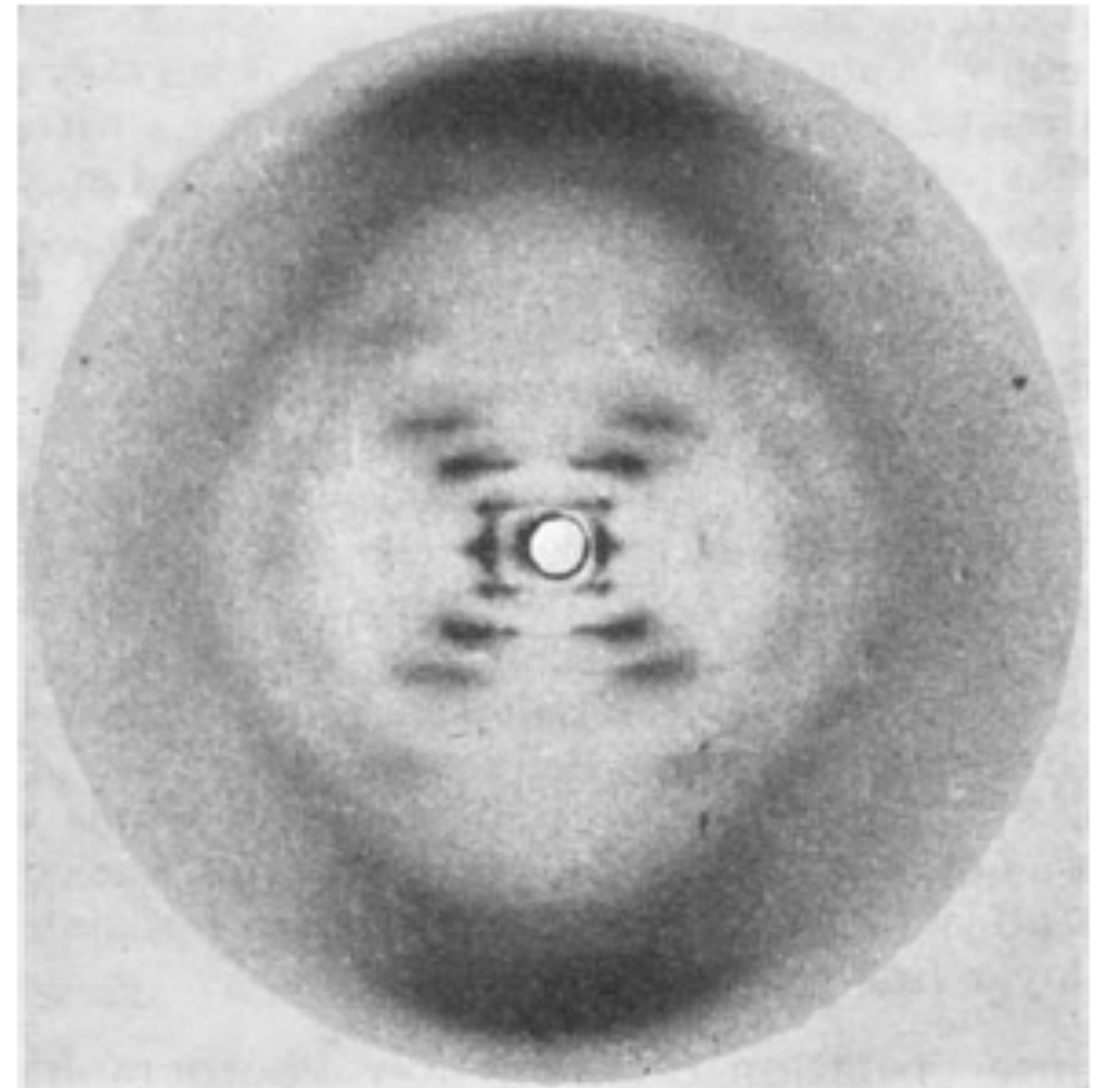
Wheatstone Physics Laboratory,
King's College, London.
April 2.

Molecular Configuration in Sodium Thymonucleate

ROSALIND E. FRANKLIN*
R. G. GOSLING

Wheatstone Physics Laboratory,
King's College, London.
April 2.

* Now at Birkbeck College Research Laboratories, 21 Torrington
Square, London, W.C.1.

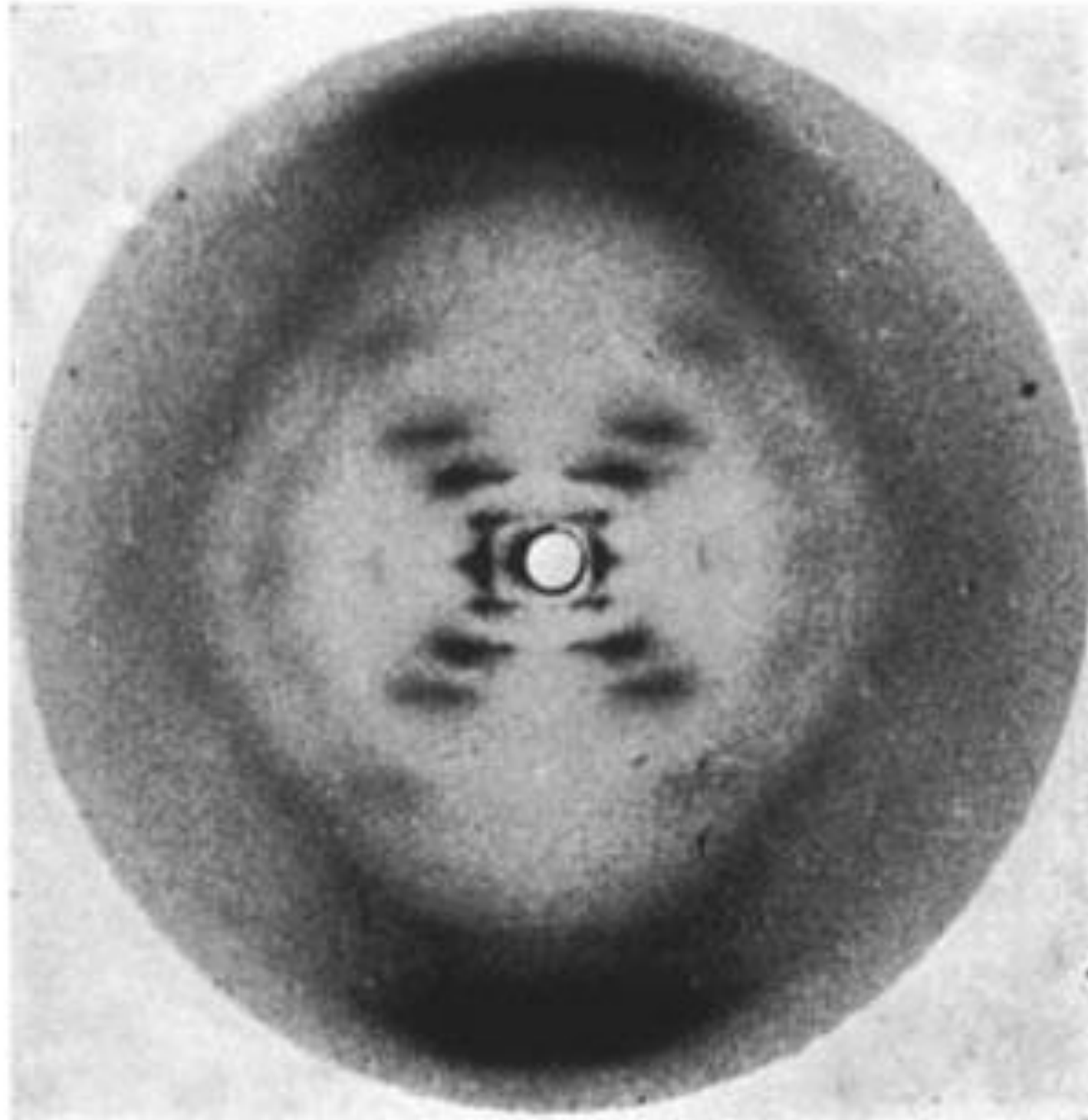


Sodium deoxyribose nucleate from calf thymus. Structure B

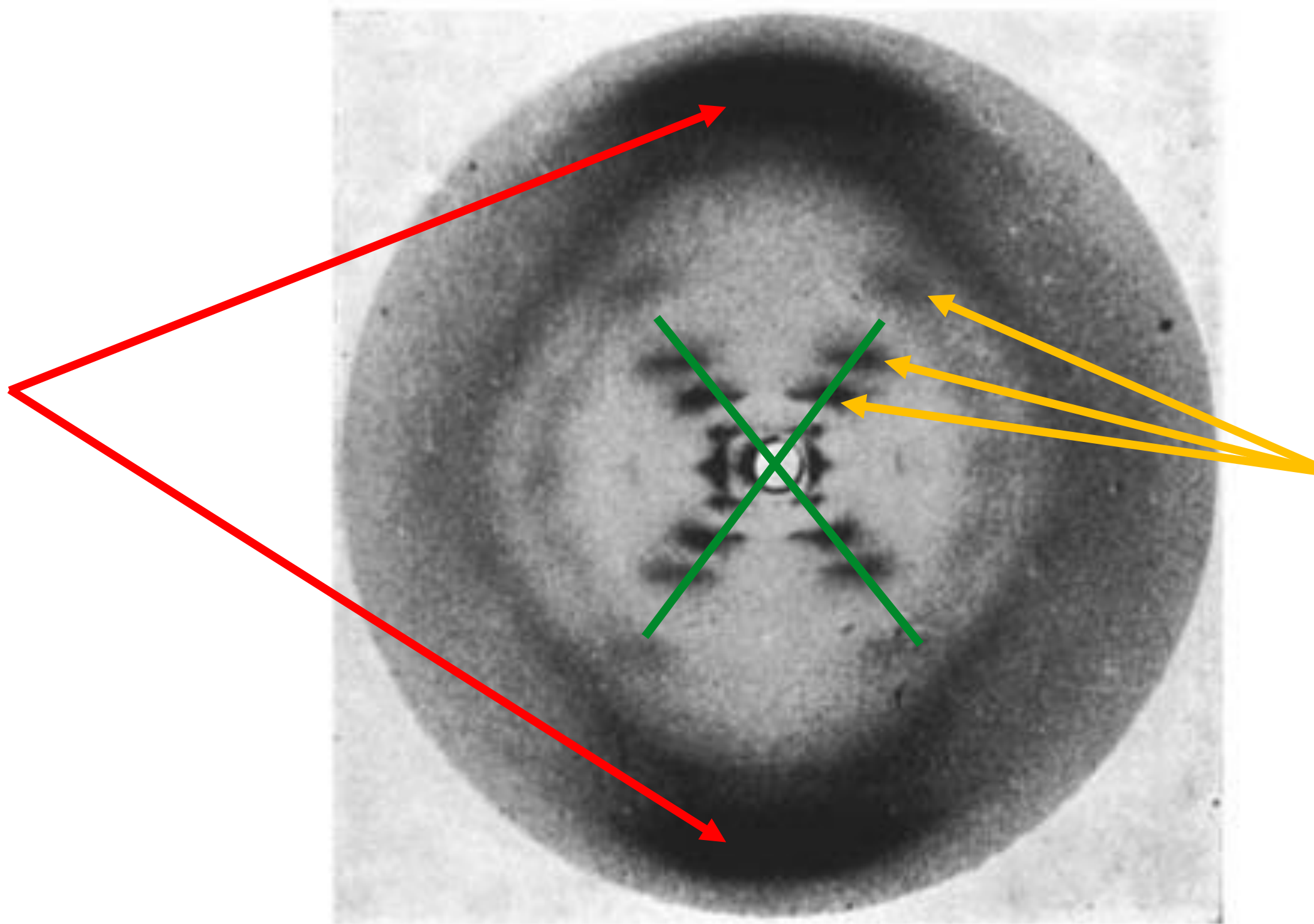
“She needed a collaborator, and she didn’t have one. Somebody to break the pattern of her thinking, to show her what was right in front of her, to push her up and over.”

– A. Klug on Franklin

Photograph 51



Standout features?



Math tool: Fourier Transform

What is a Fourier transform?

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i k x} dx$$

Output
“Frequency space”

Input
“Real space”

$$F(k_x, k_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-2\pi i (k_x x + k_y y)} dx dy$$

Sine and Cosine

(Boardwork discussion)

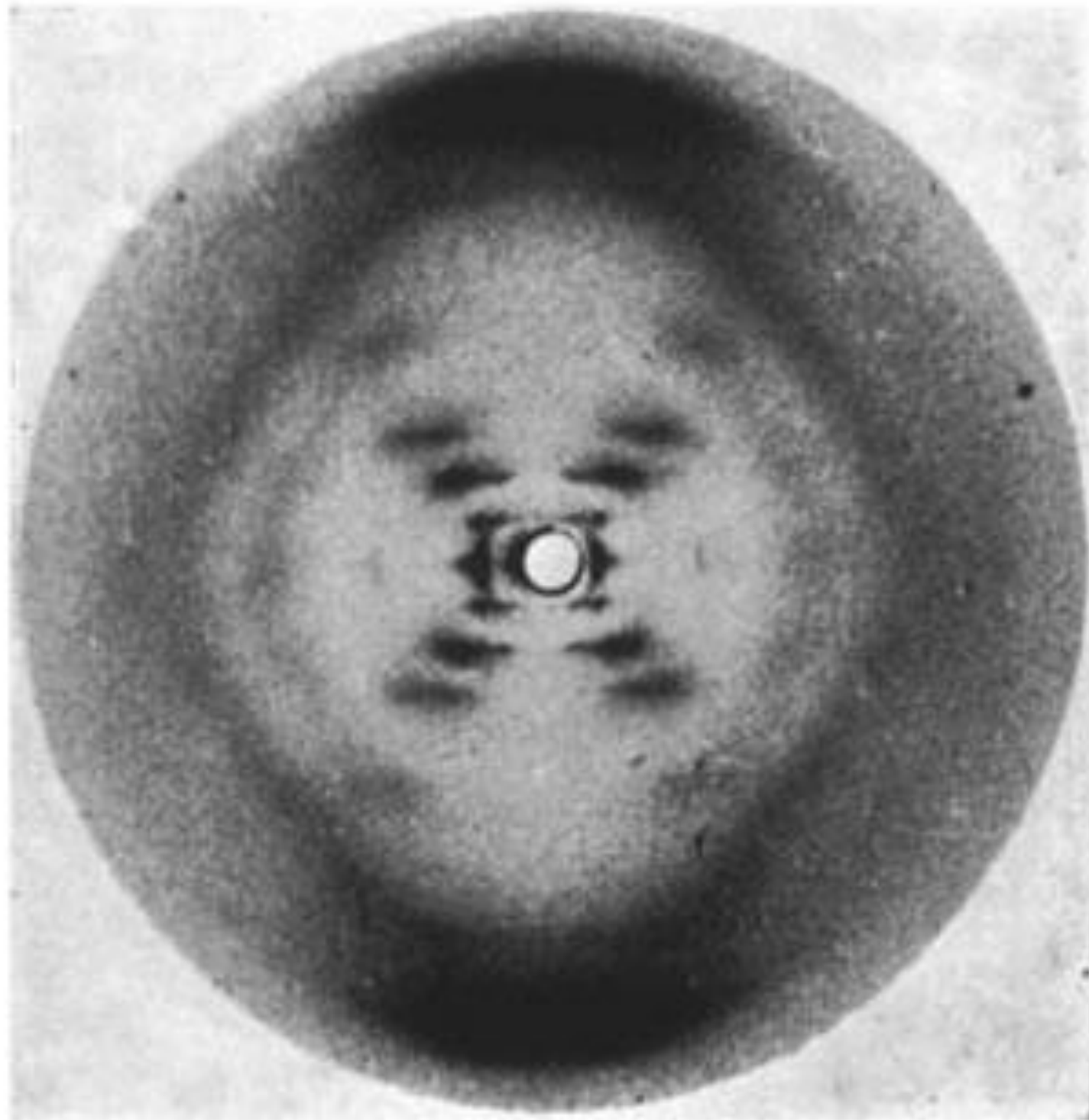
Period is inversely proportional to frequency

Fourier Transform Sine and Cosine

(Boardwork discussion)

High intensity spots signify a repeating pattern

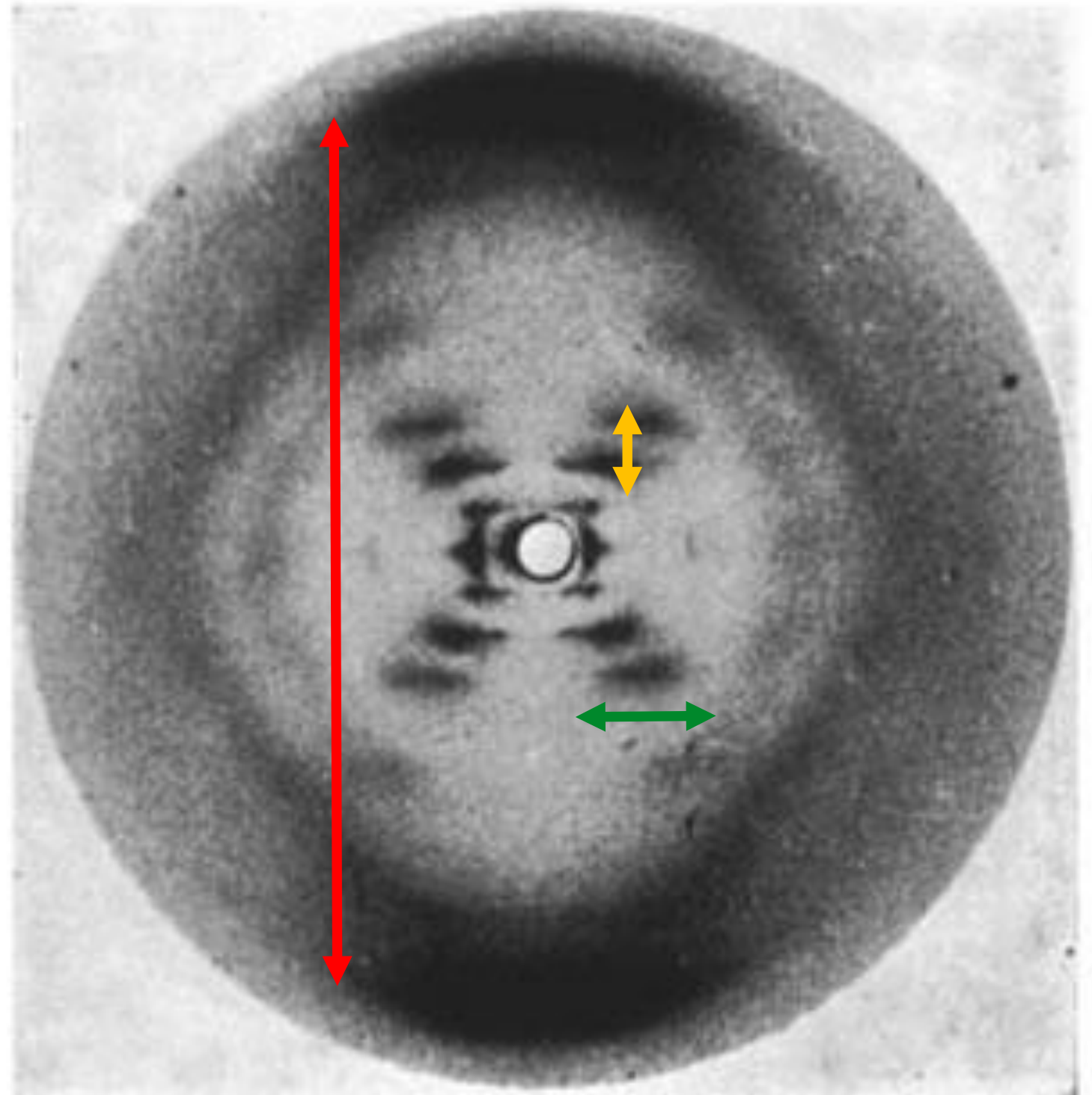
Diffraction pattern: frequency space



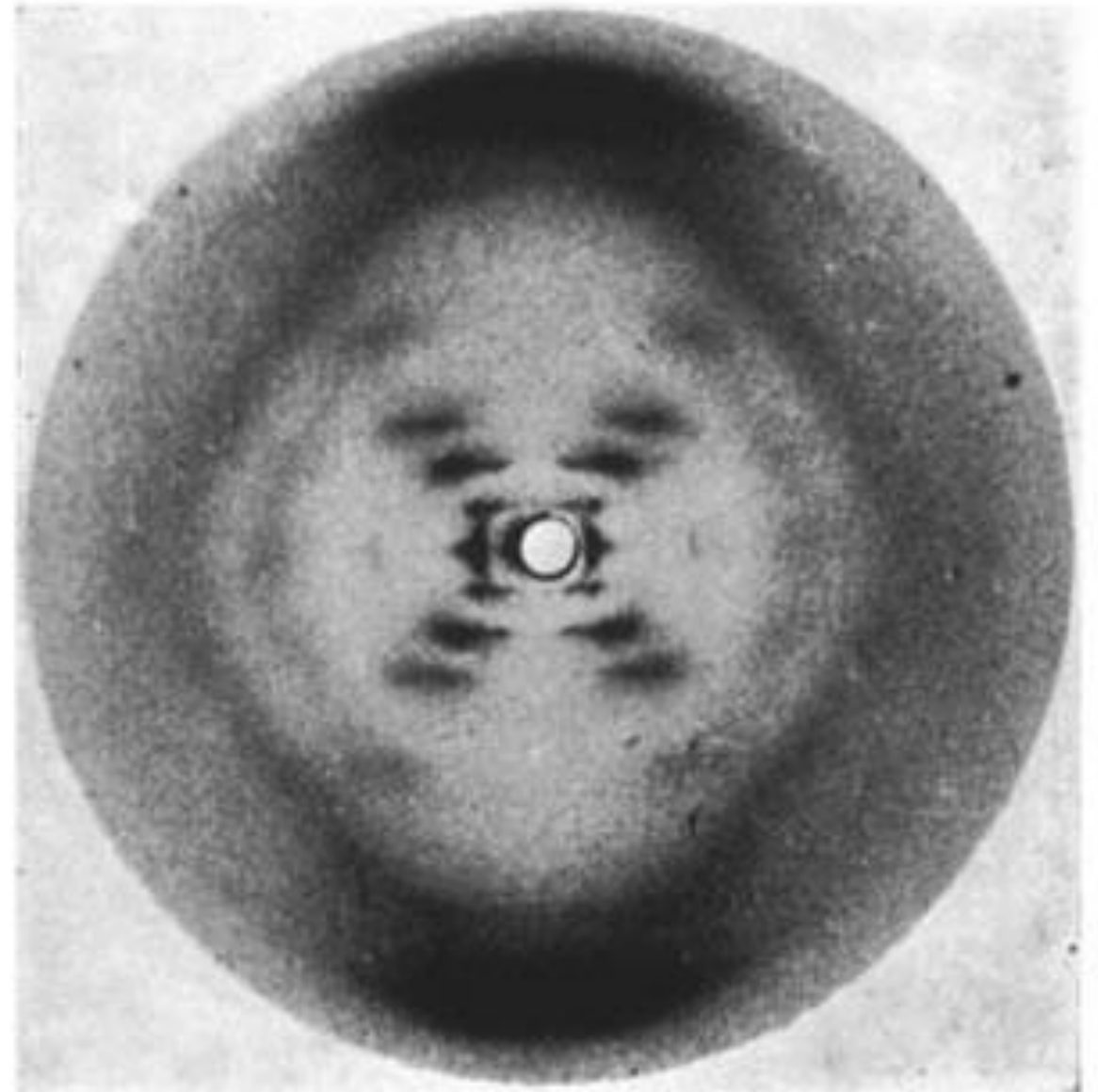
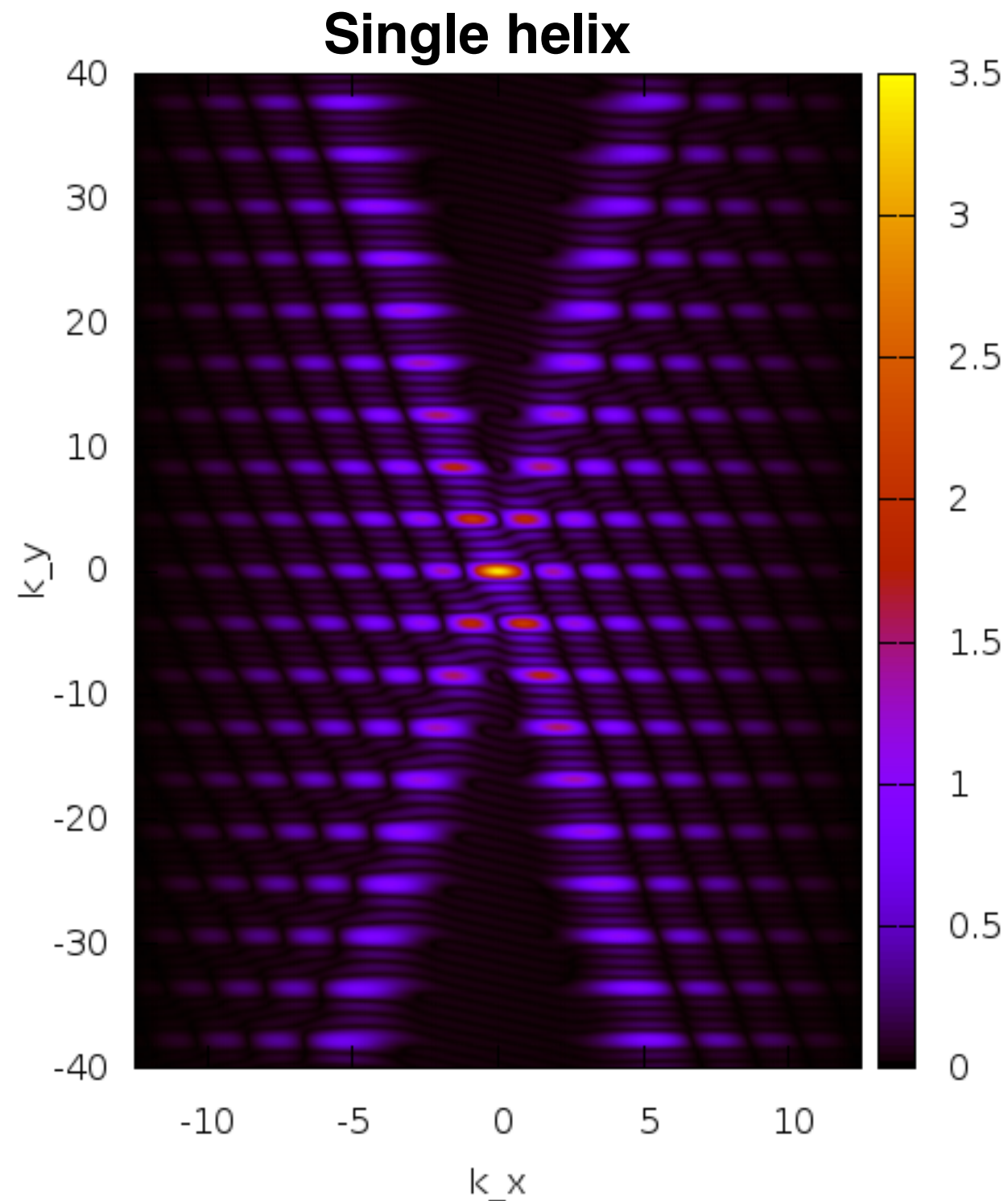
Patterns and length scales



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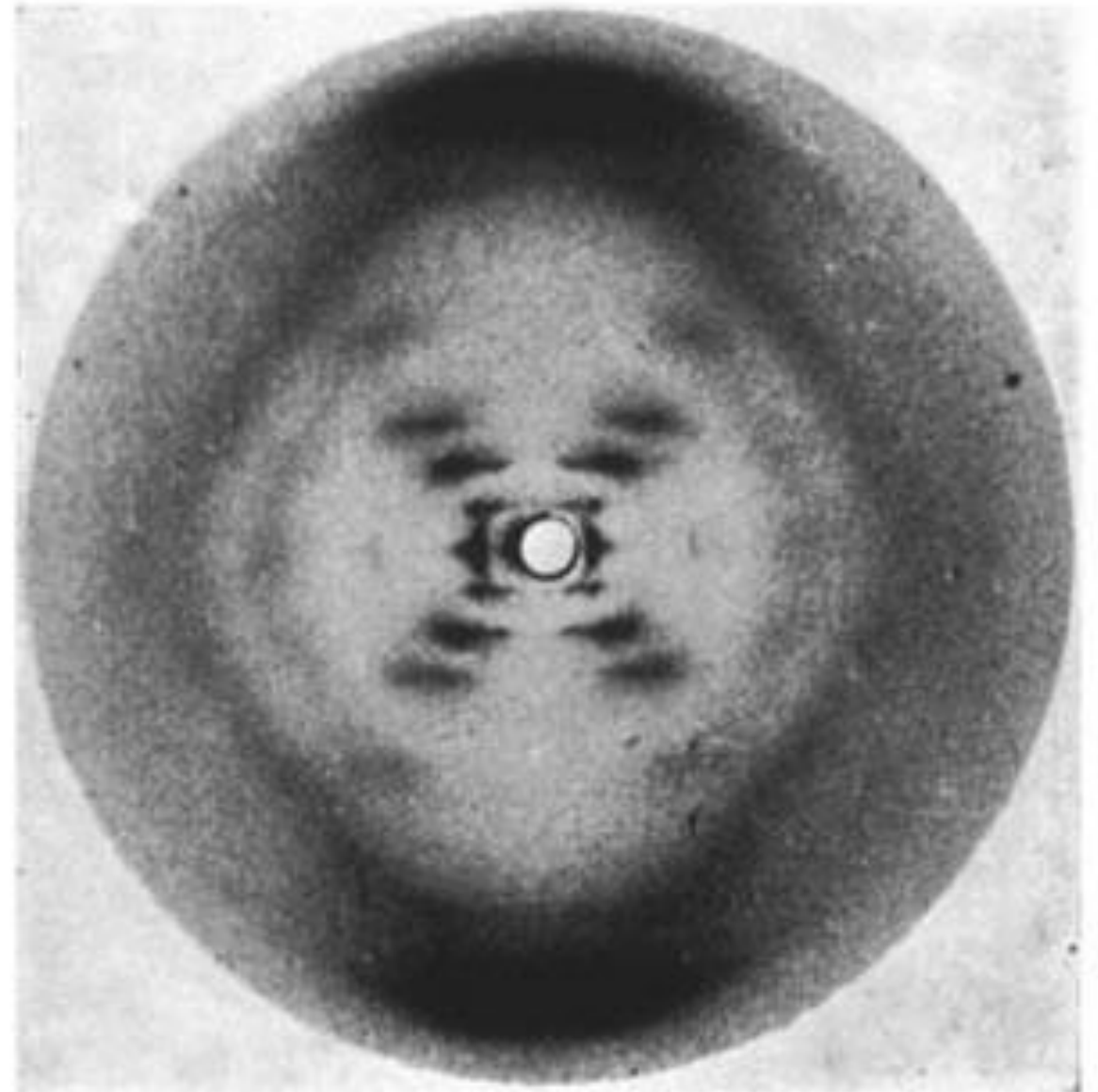
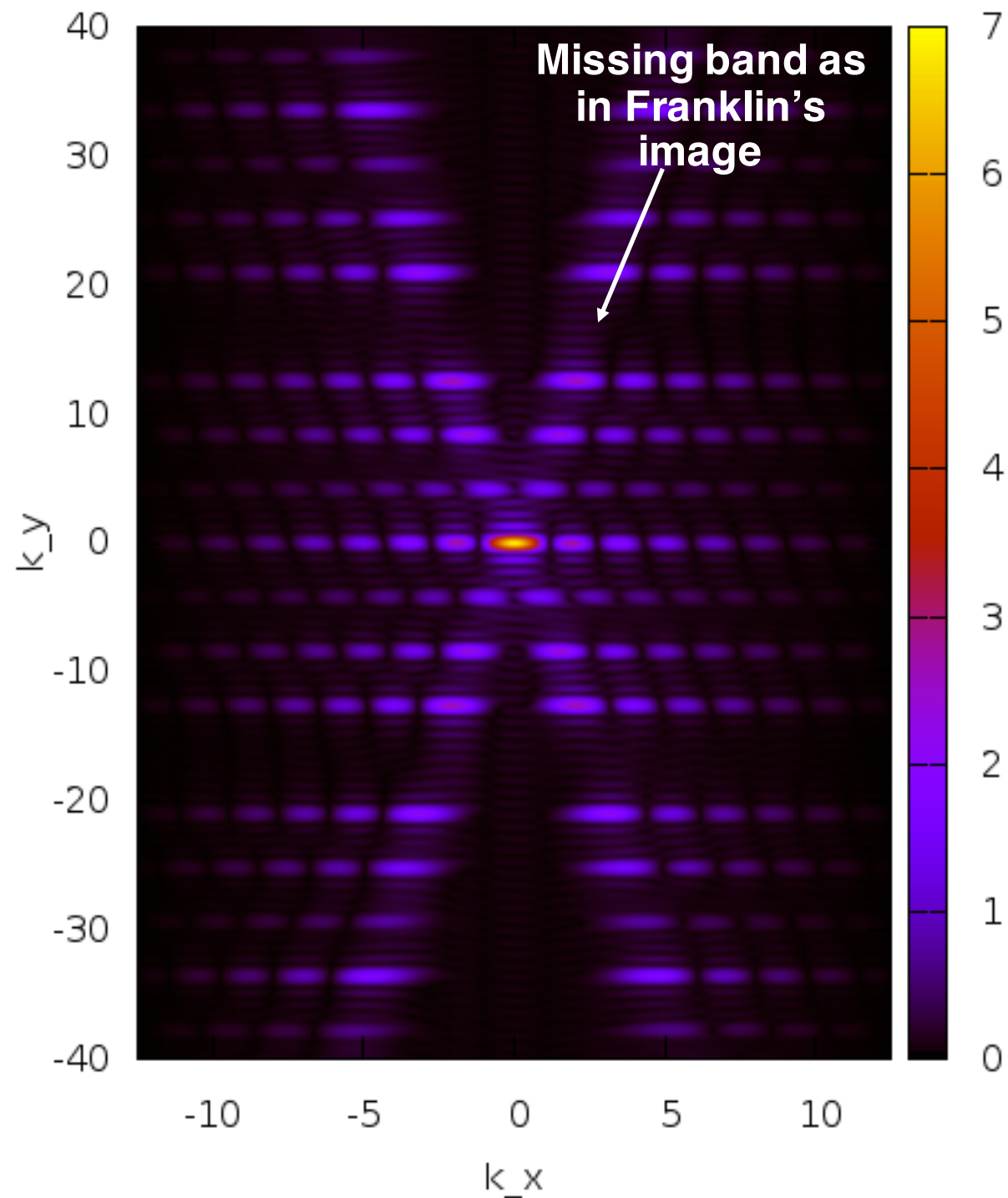


Fourier Transform: single helix



Fourier Transform: double helix

Two helices with $3/8$ displacement



The Calculation of the Century

Acta Cryst. (1952). **5**, 581

The Structure of Synthetic Polypeptides. I. The Transform of Atoms on a Helix

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The Cavendish Laboratory, Cambridge, England*

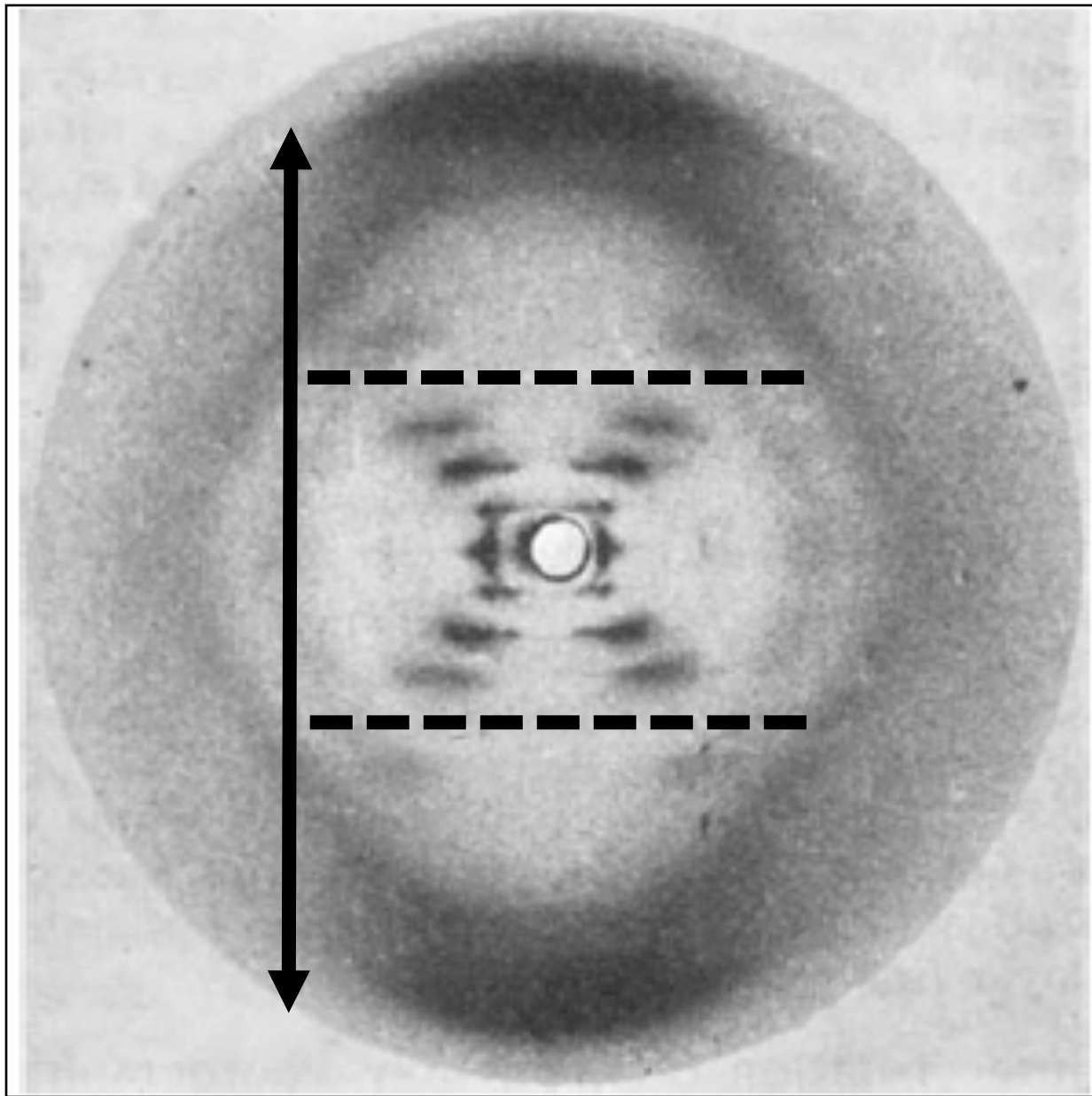
AND V. VAND*

Chemistry Department, The University, Glasgow W. 2, Scotland

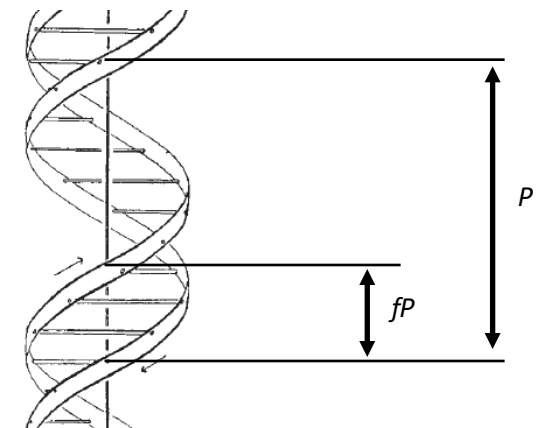
(Received 16 February 1952)

The formulæ are given for the Fourier transforms of a number of helical structures; namely, a thin helical wire, a set of identical atoms spaced at regular intervals on a helix, and the general case of a group of atoms repeated by the operation of a non-integer screw. General predictions are made concerning the intensities of the X-ray diffraction pattern of the synthetic polypeptide poly- γ -methyl-L-glutamate, assuming that its structure is based on the α -helix suggested by Pauling & Corey.

Diffraction from a double helix



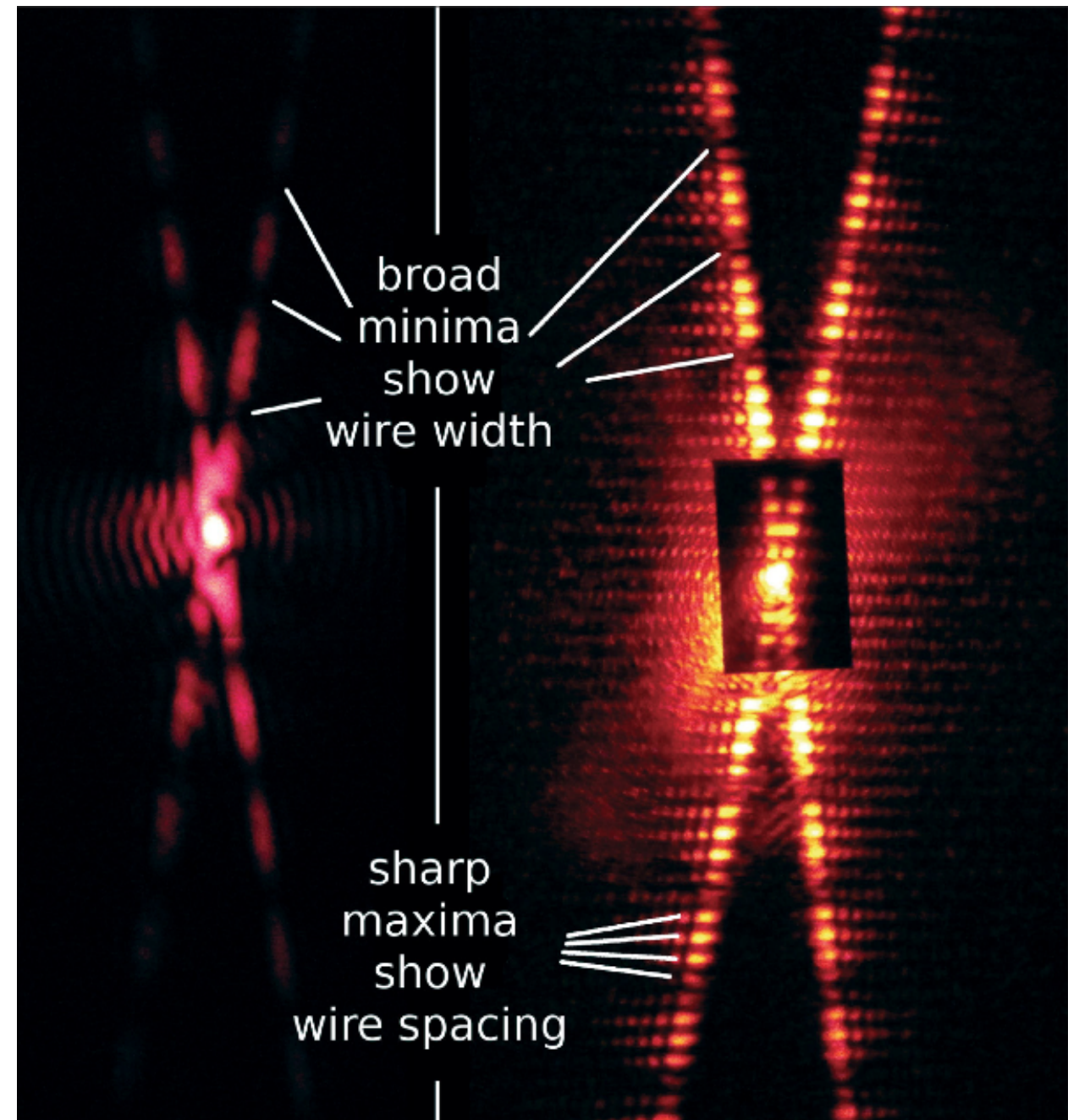
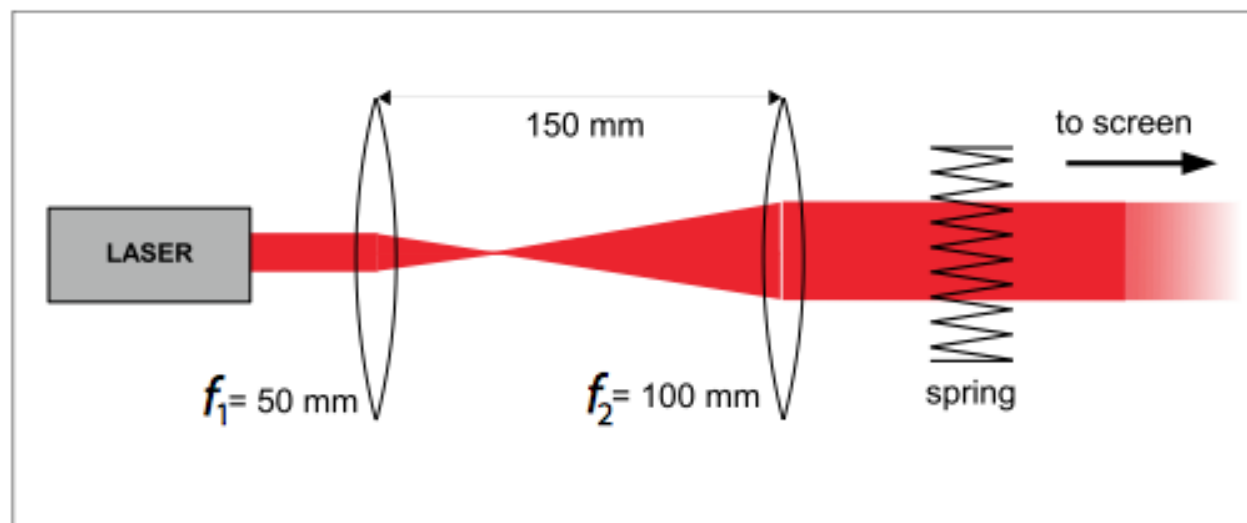
- Base pair spacing along the helix given by strong reflections at the poles.
- Two helices, each with radius R and pitch P , separated by fraction f of P . Destructive interference with $f \approx 3/8$ causes missing 4th Bessel line.



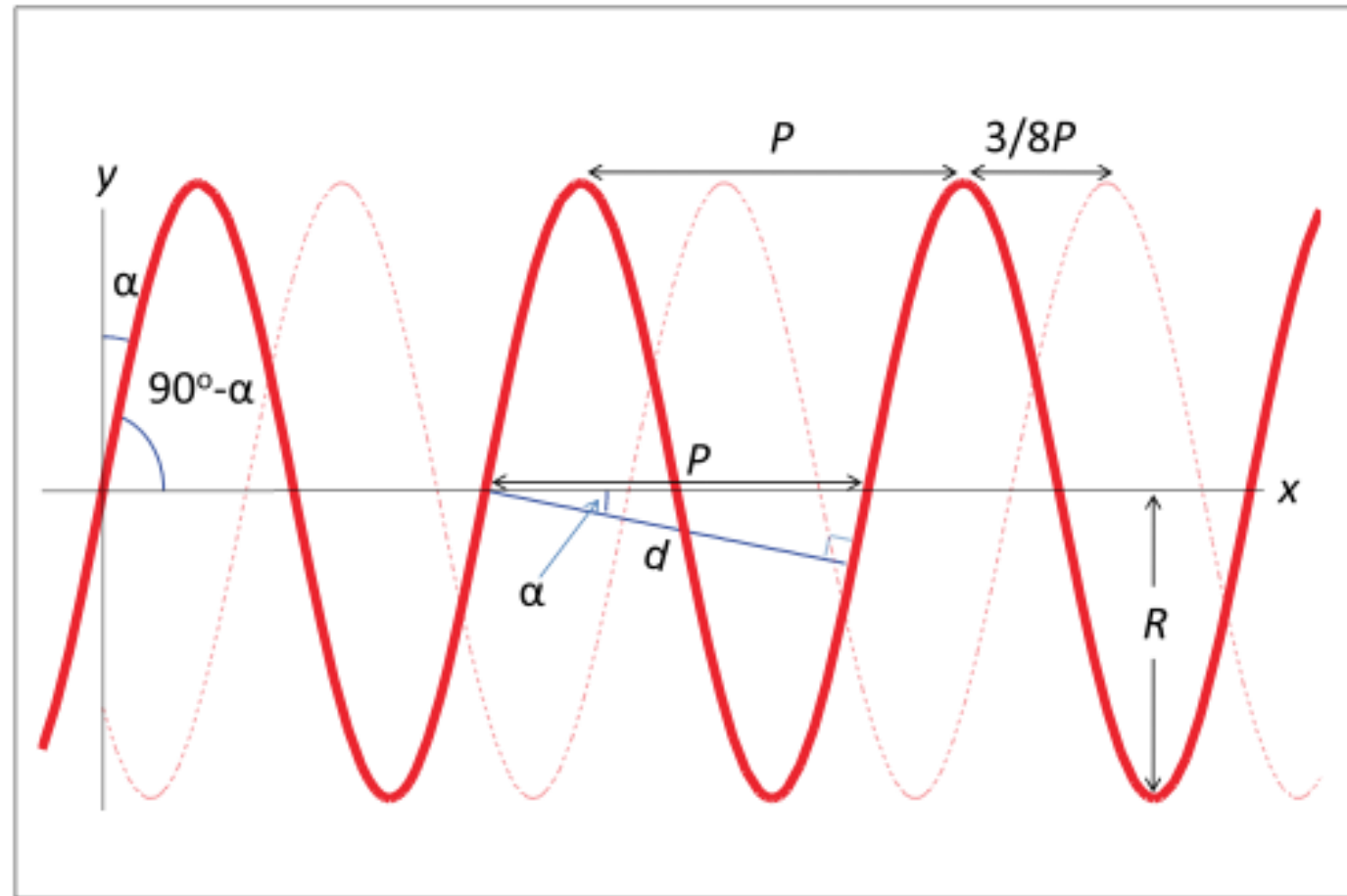
- Photo 51 gives helix pitch, angle, radius and radial mass distribution, base pair spacing, and double helix separation.

Recreation of experiment

The basic idea behind the experiment can be reproduced with a laser, beam expander, and spring from a ballpoint pen



Numerical recreation



- The diffraction pattern can also be reproduced by taking the Fourier transform of a helix, which could be approximated by

$$F(k_x, k_y) = \int_{x_1}^{x_2} \int_{R \sin\left(\frac{2\pi x}{P}\right) - \frac{a}{2}}^{R \sin\left(\frac{2\pi x}{P}\right) + \frac{a}{2}} \sin(k_x x) \sin(k_y y) dx dy$$

Thursday's class

More inverse problems

**Computed tomography and image
reconstruction**